

Abdul Khurram

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Available for Co-ops/Internships (Open to Relocation): May–Dec 2026 | Full-time: May 2027+

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Apache Spark, Scala, Airflow, Kafka, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered secure **Scala** pipelines orchestrated by **Airflow** DAGs to process **50M+ daily transactions** from **Delta Lake**, enforcing strict **RBAC/encryption** controls on **petabyte-scale** financial data.
- Optimized **Spark** jobs by migrating legacy code to **Spark-native** APIs and tuning **AQE/cluster memory**, solving resource bottlenecks and reducing **latency** by **40%**.
- Developing **multi-region** data pipelines with built-in **fault tolerance**, **idempotent** re-runs, and automated **failover**, collaborating cross-team to ship reliable **data products** for critical banking workloads.

Software Engineer Intern | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Owned **end-to-end feature delivery**: shipped scheduling UX, multi-brand management, video publishing, and Canva export — each backed by **distributed async pipelines** on **AWS (SQS/S3)** at **thousands/day** scale.
- Accelerated content prep by ingesting brand docs with an **NLP** step, caching **embeddings**, and auto-injecting channel-specific context while enforcing **API guardrails**.
- Raised reliability to **99.9%** via **OpenAI Responses API** migration, **Playwright E2E** gates in **CI/CD**, hardened auth/session, and **retry/backoff** with **CloudWatch** alerting.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led development of an **AI-driven** VR clinical simulation in Unreal (C++), collaborating with UBC's School of Nursing to train nurse practitioners.
- Designed a novel **multithreaded** supervisory architecture (C++/**Python**) that monitors **AI** patient behavior and enforces clinical protocol adherence.
- Evaluated system performance, achieving **40%** latency reduction and **90%** inference cost savings through **pipeline optimizations**.

PROJECTS

Backer | *TypeScript/Next.js, AWS (DynamoDB, S3, SES), NextAuth*

Hack The Coast 2026

- Built a **TikTok-style startup feed** with swipe interactions and watch-milestone tracking; architected a **DynamoDB-backed** data layer (**10+ tables**) with **transactional writes** powering feed, messaging, and investment workflows.
- Trained an **XGBoost ranker** scoring startups on **6 features** (stage affinity, **text similarity**, tag overlap, popularity) from **DynamoDB event streams**.

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Shipped an **end-to-end** freelancer workspace (intake, planning, proposals, delivery) with **AWS** infrastructure and **CI** gates.
- Built a user-facing email intake console that triages threads, captures scope, and renders versioned proposal PDFs via a **Vision Model**.
- Designed a **Python** evaluator→revisor **agent loop** with rubric-based **benchmarking** for measured quality improvements.

SKILLS

Languages: Java, Python, Scala, TypeScript/JavaScript, **SQL**, C/C++

Platforms & Tools: Apache Spark, Airflow, Kafka, Delta Lake, AWS, Next.js/React, Flask, Playwright, Docker, Git, RDBMS